Face Recognition using PCA and ANN

AI/ML Internship Project
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Project Overview

This project focuses on developing an intelligent face recognition system using Principal Component Analysis (PCA) and Artificial Neural Networks (ANN).

Project Objectives

- Develop an AI-based face recognition model
- Reduce image dimensions using PCA
- Train ANN for face classification
- Improve recognition efficiency and accuracy

Technologies & Tools Used

- Python
- Google Colab
- NumPy & Pandas
- Matplotlib
- Scikit-learn
- TensorFlow/Keras

System Workflow

1. Data Collection
2. Image Preprocessing
3. Feature Extraction using PCA
4. ANN Model Training
5. Face Prediction & Recognition

Principal Component Analysis (PCA)

- PCA reduces high-dimensional image data
- Extracts important facial features
- Improves computational efficiency
- Helps reduce noise and redundancy

Artificial Neural Network (ANN)

- ANN learns facial patterns from extracted features
- Performs classification and prediction
- Provides better recognition accuracy
- Mimics human brain learning process

Project Results

- Successfully implemented face recognition system
- PCA effectively reduced dimensions
- ANN achieved accurate predictions
- Model demonstrated efficient performance

Applications

- Security Systems
- Smart Attendance Systems
- Smartphone Face Unlock
- Surveillance & Monitoring
- Identity Verification

Conclusion

The project successfully demonstrates the practical application of AI and Machine Learning in face recognition using PCA and ANN techniques.